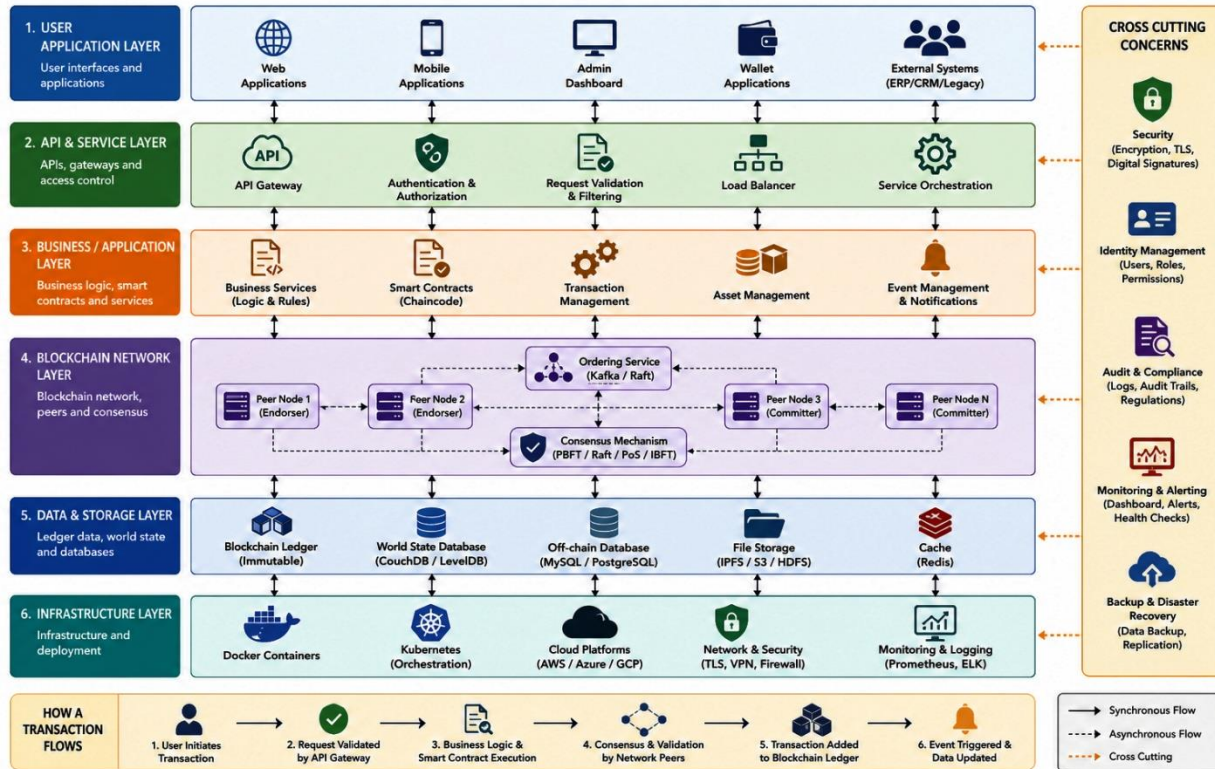
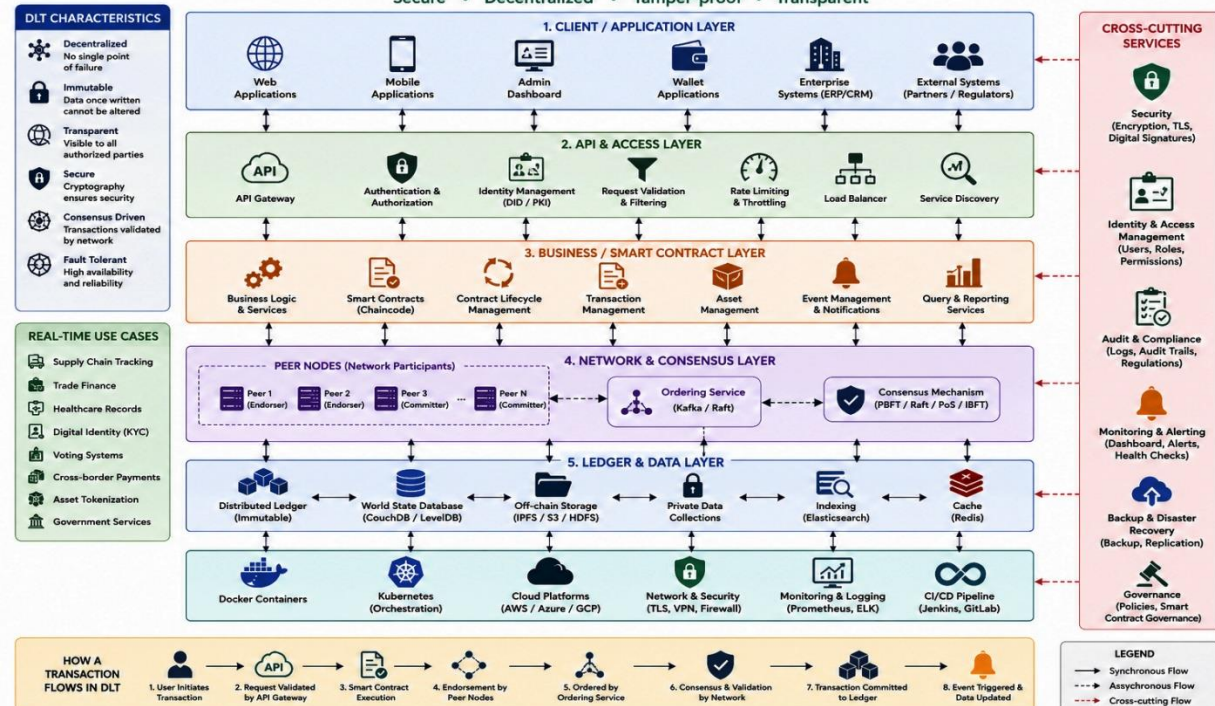


BLOCKCHAIN ARCHITECTURE



DISTRIBUTED LEDGER TECHNOLOGY (DLT) ARCHITECTURE

Secure • Decentralized • Tamper-proof • Transparent



Below is a comprehensive table of the major technologies, frameworks, tools, platforms, databases, programming languages, DevOps tools, and cloud services commonly used in **Blockchain / Distributed Ledger Technology (DLT)** projects, along with their purpose, real-time use cases, and how to install them on a local laptop.

Technology	Category	Description	Real-Time Example	Used By	Local Installation
Bitcoin	Public Blockchain	World's first decentralized blockchain for digital currency	Cryptocurrency payments	Crypto Exchanges	Install Bitcoin Core
Ethereum	Public Blockchain	Smart contract platform supporting DApps	NFT Marketplace	DeFi Companies	Install Geth
Hyperledger Fabric	Permissioned Blockchain	Enterprise blockchain with private channels	Supply Chain	IBM, Walmart	Install Docker + Fabric Samples
Hyperledger Sawtooth	Permissioned Blockchain	Modular enterprise blockchain	Healthcare	Intel	Install Docker images
Hyperledger Besu	Enterprise Ethereum	Enterprise Ethereum client	Banking	ConsenSys	Install Besu JAR
Hyperledger Indy	Identity Blockchain	Decentralized Identity (DID)	Digital Identity	Government	Install Indy Node
Hyperledger Iroha	Enterprise Blockchain	Mobile-friendly blockchain	Banking	Financial Organizations	Install Docker
Hyperledger Burrow	Ethereum Smart Contract	Permissioned Ethereum implementation	Enterprise Smart Contracts	Enterprises	Install Go binaries

Corda	DLT	Financial blockchain platform	Trade Finance	Banks	Install Corda Node
Quorum	Enterprise Ethereum	Permissioned Ethereum	Interbank Settlement	JP Morgan	Install Quorum binaries
Polygon	Layer-2	Ethereum scaling solution	NFT Marketplace	Polygon Labs	Install Polygon Edge
Solana	High-Speed Blockchain	Fast blockchain network	Web3 Gaming	Solana Labs	Install Solana CLI
Avalanche	Blockchain Platform	High throughput blockchain	DeFi	Ava Labs	Install AvalancheGo
Cardano	Blockchain	Proof-of-Stake blockchain	Education Certificates	IOHK	Install Cardano Node
Polkadot	Multi-chain	Connects multiple blockchains	Cross-chain Assets	Web3 Foundation	Install Polkadot Node
Cosmos	Blockchain Network	Internet of Blockchains	Multi-chain Communication	Cosmos	Install Gaia
Ripple XRP Ledger	Payment Blockchain	Fast international payment network	Cross-border Payments	Banks	Install rippled
Stellar	Payment Blockchain	Financial inclusion blockchain	Money Transfer	NGOs	Install Stellar Core
Tron	Blockchain	High TPS blockchain	Entertainment	Tron Foundation	Install Tron Node
EOS	Blockchain	High performance blockchain	Gaming	EOS Network	Install EOSIO

Smart Contract Technologies

Technology	Description	Language	Real-Time Example	Installation
Solidity	Ethereum Smart Contract Language	Solidity	ERC20 Token	Install Node.js then npm install solc
Vyper	Python-like Smart Contract	Python-like	DeFi Projects	pip install viper
Rust	Solana Smart Contracts	Rust	Solana Programs	Install Rust
Move	Aptos/Sui Smart Contracts	Move	Digital Assets	Install Move CLI
DAML	Financial Smart Contracts	DAML	Banking Contracts	Install DAML SDK
Chaincode	Hyperledger Fabric Smart Contracts	Java, Go, Node.js	Supply Chain	Fabric Samples

Programming Languages

Language	Used For	Real-Time Example	Installation
Java	Hyperledger Fabric Chaincode	Supply Chain	Install JDK 8/11/17
Go	Fabric Chaincode	Banking	Install Go
Node.js	Smart Contracts	NFT Marketplace	Install Node.js
Python	Automation	Blockchain Monitoring	Install Python
JavaScript	DApps	Wallet UI	Install Node.js
TypeScript	Enterprise DApps	Web3 Apps	npm install typescript
C++	Bitcoin Core	Cryptocurrency	Install Visual Studio

Rust	Solana Programs	Web3 Gaming	Install Rustup
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Databases

Database	Purpose	Example	Installation
CouchDB	World State	Hyperledger Fabric	Install CouchDB
LevelDB	Default Ledger DB	Hyperledger Fabric	Included
RocksDB	High Performance	Besu	Install RocksDB
PostgreSQL	Off-chain Storage	Banking	Install PostgreSQL
MySQL	User Data	Wallet Application	Install MySQL
MongoDB	Metadata	NFT Platform	Install MongoDB
Redis	Cache	Wallet API	Install Redis

Wallet Technologies

Wallet	Description	Example	Installation
MetaMask	Ethereum Wallet	DApps	Browser Extension
Trust Wallet	Mobile Wallet	Crypto Payments	Mobile App
Phantom	Solana Wallet	Solana DApps	Browser Extension
Coinbase Wallet	Web3 Wallet	NFT	Browser Extension
Ledger Nano	Hardware Wallet	Asset Storage	Ledger Live
Trezor	Hardware Wallet	Cold Wallet	Install Trezor Suite

Consensus Algorithms

Algorithm	Description	Example
Proof of Work (PoW)	Mining-based	Bitcoin

Proof of Stake (PoS)	Validators	Ethereum 2.0
Delegated PoS	Delegated Validators	EOS
Practical Byzantine Fault Tolerance (PBFT)	Enterprise	Hyperledger
RAFT	Crash Fault Tolerance	Quorum
IBFT	Byzantine Consensus	Besu
Proof of Authority	Trusted Validators	Enterprise Networks

DevOps Tools

Tool	Purpose	Example	Installation
Docker	Containers	Fabric Network	Install Docker Desktop
Kubernetes	Container Orchestration	Production Cluster	Install Minikube
Helm	Kubernetes Packages	Fabric Charts	Install Helm
Jenkins	CI/CD	Smart Contract Deployment	Install Jenkins
Git	Version Control	Source Code	Install Git
GitHub	Repository	Blockchain Code	GitHub Desktop
GitLab	DevOps	Enterprise CI/CD	GitLab CE

Cloud Platforms

Cloud	Purpose	Example
AWS	Blockchain Hosting	Managed Blockchain
Azure	Blockchain Infrastructure	Azure Kubernetes
Google Cloud	Blockchain Nodes	GKE

Oracle Cloud	Enterprise Blockchain	OCI
IBM Cloud	Hyperledger Hosting	IBM Blockchain

Blockchain Frameworks

Framework	Purpose	Real-Time Example
Truffle	Smart Contract Development	Ethereum
Hardhat	Smart Contract Testing	DeFi
Brownie	Python Ethereum Framework	Automation
Foundry	Solidity Development	High-speed Testing
Remix IDE	Online Solidity IDE	Learning

APIs

API	Purpose	Example
Web3.js	JavaScript Blockchain API	Wallet Connection
ethers.js	Ethereum Library	Smart Contracts
Fabric SDK Java	Hyperledger	Enterprise Apps
Fabric SDK Node	Hyperledger	REST APIs
JSON RPC	Ethereum Communication	Node Access

Identity Technologies

Technology	Purpose	Example
DID	Decentralized Identity	Digital Passport
Verifiable Credentials	Digital Certificates	University Degree
Hyperledger Aries	Identity Agent	Digital Wallet

Hyperledger Indy	Identity Ledger	Government IDs
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Monitoring Tools

Tool	Purpose	Example
Prometheus	Metrics	Blockchain Monitoring
Grafana	Dashboard	Node Health
ELK Stack	Logs	Smart Contract Logs
Kibana	Visualization	Blockchain Analytics

Security Tools

Tool	Purpose	Example
HashiCorp Vault	Secrets Management	Private Keys
OpenSSL	Certificates	TLS
SonarQube	Code Quality	Smart Contract Scan
Trivy	Vulnerability Scanner	Docker Images

Message Brokers

Tool	Purpose	Example
Kafka	Event Streaming	Supply Chain Events
RabbitMQ	Messaging	Blockchain Notifications
ActiveMQ	Enterprise Messaging	Banking

Local Laptop Installation Order (Recommended)

Step	Software	Why Required
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1	Git	Source code management
2	Java JDK 8/11/17	Java development
3	Maven	Build Java projects
4	Node.js & npm	Smart contracts and DApps
5	Go	Hyperledger Fabric chaincode
6	Python 3	Automation and scripts
7	Docker Desktop	Run blockchain nodes in containers
8	Docker Compose	Multi-container blockchain networks
9	Kubernetes (Minikube/Kind)	Local orchestration
10	Visual Studio Code	Code editor
11	Hyperledger Fabric Samples	Enterprise blockchain development
12	CouchDB	Fabric world state database
13	MySQL/PostgreSQL	Off-chain application data
14	Postman	Test REST APIs
15	MetaMask	Ethereum wallet for DApps
16	Truffle or Hardhat	Smart contract development
17	Ganache	Local Ethereum blockchain
18	Prometheus & Grafana	Monitoring
19	Jenkins	CI/CD pipelines
20	Helm	Kubernetes deployments

Complete Enterprise Blockchain Stack Example (Supply Chain Management)

Layer	Technologies
Frontend	React, Angular, Vue.js
Backend	Java 8+, Spring Boot, REST API

Blockchain	Hyperledger Fabric
Smart Contracts	Java Chaincode / Go Chaincode
Database	CouchDB, MySQL
Messaging	Kafka
Security	TLS, OpenSSL, HashiCorp Vault
Authentication	JWT, OAuth2, LDAP
Containerization	Docker
Orchestration	Kubernetes
CI/CD	Jenkins, GitHub Actions
Monitoring	Prometheus, Grafana
Logging	ELK Stack
Cloud	AWS, Azure, GCP
Build Tool	Maven, Gradle
API Testing	Postman
IDE	IntelliJ IDEA, VS Code
Version Control	Git, GitHub

This stack represents a modern enterprise-grade blockchain architecture suitable for banking, supply chain, healthcare, trade finance, identity management, and other large-scale distributed ledger applications.